01_data_discovery_series_exploration

July 21, 2026

1 Bank of Canada Valet API - Series Catalog Exploration

1.1 Objective

The goal of this notebook is to explore the Bank of Canada Valet API series catalog.

This first exploration phase aims to:

- Understand the structure of the available financial series
- Analyze metadata quality
- Identify relevant economic and financial indicators
- Select candidate series for the Canadian Financial Lakehouse project

The dataset contains more than 15,000 available time series.

[5]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path



pd.set_option("display.max_columns", None)
pd.set_option("display.max_colwidth", 200)
```

1.2 Load Series Catalog

The series catalog was extracted from the Bank of Canada Valet API.

Each record represents one available economic or financial time series.

Main attributes:

- `series_id`: Unique identifier of the series
- `label`: Short title of the series
- `description`: Detailed description
- `link`: API documentation URL

[3]:
```
catalog_path = "../discovery/outputs/series_catalog.csv"

df = pd.read_csv(catalog_path)
```

```
df.head()
```

[3]:
```
  series_id                          label  \
0    A.AGRI       Annual BCPI Agriculture
1    A.BCNE  Annual BCPI Excluding Energy
2    A.BCPI             Annual BCPI Total
3    A.ENER            Annual BCPI Energy
4    A.FISH              Annual BCPI Fish

                                        description  \
0  Annual Bank of Canada commodity price index - …
1  Annual Bank of Canada commodity price index - …
2  Annual Bank of Canada commodity price index - …
3  Annual Bank of Canada commodity price index - …
4  Annual Bank of Canada commodity price index - …

                                               link
0  https://www.bankofcanada.ca/valet/series/A.AGRI
1  https://www.bankofcanada.ca/valet/series/A.BCNE
2  https://www.bankofcanada.ca/valet/series/A.BCPI
3  https://www.bankofcanada.ca/valet/series/A.ENER
4  https://www.bankofcanada.ca/valet/series/A.FISH
```

1.3 Dataset Overview

Before selecting financial indicators, we need to understand the size and structure of the catalog.

This step helps identify:

- Number of available series
- Available metadata fields
- Potential missing information

[6]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15906 entries, 0 to 15905
Data columns (total 4 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   series_id    15906 non-null  str
 1   label        15904 non-null  str
 2   description  10794 non-null  str
 3   link         15906 non-null  str
dtypes: str(4)
memory usage: 497.2 KB
```

1.4 Initial Observation

The catalog contains:

- 15,906 financial and economic series
- 4 metadata columns
- Complete identifiers and API links
- Some missing descriptions

The `series_id` column will be used as the main reference key during the ingestion phase.

The missing descriptions will be analyzed before applying keyword-based discovery methods.

1.5 Missing Values Analysis

Understanding missing values is an important step in the data discovery phase.

Before using the catalog for automatic series selection, we need to evaluate the completeness of the metadata.

This analysis focuses on:

- Missing values by column
- Data availability for descriptions and labels
- Potential impact on the discovery process

[7]:
```
missing_values = df.isnull().sum()

missing_values
```

[7]:
```
series_id         0
label             2
description    5112
link              0
dtype: int64
```

1.6 Missing Values Percentage

To better understand the impact of missing information, we calculate the percentage of missing values for each column.

This metric provides a normalized view of data completeness regardless of dataset size.

[8]:
```
missing_percentage = (
    df.isnull().sum()
    / len(df)
    * 100
)

missing_percentage.round(2)
```

[8]:
```
series_id       0.00
label           0.01
description    32.14
link            0.00
```

```
dtype: float64
```

1.7 Missing Values Observation

The missing values analysis shows that:

- `series_id` and `link` columns are complete with no missing records.
- The `label` column contains only 2 missing values, which has a negligible impact.
- The `description` column contains 5,112 missing values.

The missing descriptions represent approximately one third of the catalog.

Since the `description` field is important for keyword-based financial discovery, we will create additional exploration methods using both `label` and `description` fields.

2 Exploring Series Labels and Descriptions

The Bank of Canada catalog contains thousands of economic and financial indicators.

Before selecting specific datasets, we need to understand the terminology used in the series metadata.

This exploration will help identify:

- Main financial domains
- Common keywords
- Potential indicators for the Canadian Financial Lakehouse

[9]:
```
df[['series_id', 'label', 'description']].head(20)
```

[9]:

	series_id
0	A.AGRI
1	A.BCNE
2	A.BCPI
3	A.ENER
4	A.FISH
5	A.FOPR
6	A.MTLS
7	ABOVE3
8	ACTUAL
9	AR_2023_C1_S1
10	AR_2023_C1_S2
11	AR_2023_C1_S3
12	AR_2024_C1_S1
13	AR_2024_C1_S2
14	AR_2024_C1_S3
15	ATOM_V41693242
16	AUC_BOND_ALLOTMENT_RATIO
17	AUC_BOND_AMOUNT
18	AUC_BOND_AUCTION_DATE

```
19        AUC_BOND_AVG_PRICE

                                                label  \
0                                Annual BCPI Agriculture
1                             Annual BCPI Excluding Energy
2                                   Annual BCPI Total
3                                  Annual BCPI Energy
4                                    Annual BCPI Fish
5                                 Annual BCPI Forestry
6                           Annual BCPI Metals and Minerals
7                             Above 3 per cent (% of firms)
8                                                ACTUAL
9                                                  2023
10                                                 2022
11                                      2011-21 average
12                                                 2024
13                                                 2023
14                                      2012-22 average
15  CPIX, percentage change year-over-year (unadjusted)
16                                    Allotment ratio (%)
17                                            Amount ($M)
18                                           Auction date
19                                              Avg price

                                   description
0
Annual Bank of Canada commodity price index - Agriculture
1
Annual Bank of Canada commodity price index - Excluding Energy
2
Annual Bank of Canada commodity price index - Total
3
Annual Bank of Canada commodity price index - Energy
4
Annual Bank of Canada commodity price index - Fish
5
Annual Bank of Canada commodity price index - Forestry
6
Annual Bank of Canada commodity price index - Metals and Minerals
7                                                                    Inflation
Expectations - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
8
Lynx settlement balances - actual
9
10
11
```

```
12
NaN
13
NaN
14
NaN
15  A measure of core inflation that excludes eight of the most volatile
Consumer Price Index (CPI) components - fruit, vegetables, gasoline, fuel oil,
natural gas, mortgage interest, inter-city trans…
16
Allotment ratio (%)
17
Amount ($M)
18
Auction date
19
Avg price
```

2.1 Initial Series Catalog Observations

The first records show that the Bank of Canada catalog contains different categories of financial and economic information.

Observed categories include:

- Commodity price indexes (BCPI)
- Inflation indicators
- Core inflation measures
- Bond auction information
- Settlement and operational data

Some series are highly relevant for economic analysis, while others are operational or administrative datasets.

A filtering strategy will be required to identify the most valuable indicators for the Canadian Financial Lakehouse.

2.2 Analyze Series Identifier Structure

The `series_id` field appears to follow a naming convention.

The first part of the identifier may indicate the observation frequency.

Examples:

- `A.BCPI` → Annual commodity price index
- `M.xxx` → Monthly series
- `D.xxx` → Daily series

We will extract this information to better understand the catalog structure.

[11]:
```
df["frequency"] = (
    df["series_id"]
    .str.split(".")
    .str[0]
)

df["frequency"].value_counts().head(20)
```

[11]:
```
frequency
WGTS         9
A            7
BD           7
M            7
W            7
TB           5
CDN          4
CP           3
BA           2
EUROCAE01    2
IEXE0301     2
IEXE0701     2
IEXE0901     2
IEXE1001     2
IEXE1101     2
IEXE1201     2
IEXE1401     2
IEXE1601     2
IEXE1901     2
IEXE2001     2
Name: count, dtype: int64
```

2.3 Series Identifier Structure Observation

The initial analysis shows that the `series_id` naming convention is not fully standardized.

Some identifiers contain frequency information:

- `A.xxx` appears to represent annual series
- `M.xxx` appears to represent monthly series
- `D.xxx` appears to represent daily series

However, many identifiers follow different naming patterns.

Therefore, frequency information should not be extracted only from the `series_id` field.

Additional metadata exploration will be required to correctly classify the available series.

[13]:
```
common_frequency = (
    df["frequency"]
    .isin(
```

```
        [
            "A",
            "M",
            "Q",
            "D",
            "W"
        ]
    )
)

df[common_frequency]["frequency"].value_counts()
```

[13]:
```
frequency
A    7
M    7
W    7
Name: count, dtype: int64
```

2.4 Frequency Extraction Observation

The analysis shows that only a small number of series follow a simple frequency prefix convention.

Detected standard prefixes:

- `A` : Annual
- `M` : Monthly
- `W` : Weekly

However, this pattern applies to a very limited number of records.

Most Bank of Canada series use other identifier formats, meaning that frequency information cannot reliably be derived from the `series_id` alone.

For the next steps, frequency metadata should be retrieved from the API response or extracted from the original series metadata.

2.5 Prepare Text Fields for Search

The catalog contains missing descriptions.

To perform keyword-based exploration, we create a cleaned text field by combining:

- Series label
- Series description

This allows us to search across all available metadata.

[14]:
```
df["search_text"] = (
    df["label"]
    .fillna("")
    + " "
    + df["description"]
```

```
    .fillna("")
)

df[["series_id", "search_text"]].head()
```

[14]:
```
  series_id  \
0    A.AGRI
1    A.BCNE
2    A.BCPI
3    A.ENER
4    A.FISH

   search_text
0            Annual BCPI Agriculture Annual Bank of Canada commodity price index
- Agriculture
1  Annual BCPI Excluding Energy Annual Bank of Canada commodity price index -
Excluding Energy
2                       Annual BCPI Total Annual Bank of Canada commodity price
index - Total
3                     Annual BCPI Energy Annual Bank of Canada commodity price
index - Energy
4                         Annual BCPI Fish Annual Bank of Canada commodity
price index - Fish
```

2.6 Text Preparation Observation

The new `search_text` field combines available information from both labels and descriptions.

This approach improves the discovery process because:

- Some series have detailed descriptions but short labels.
- Some series only provide meaningful information in the label.
- Missing descriptions are handled automatically.

The combined text field will be used for keyword-based financial indicator discovery.

3 Keyword-Based Series Search

To identify relevant datasets for the Canadian Financial Lakehouse, we will search the catalog using financial and economic keywords.

This function allows us to quickly explore available indicators related to specific domains.

[15]:
```
def search_series(keyword):

    result = df[
        df["search_text"]
        .str.contains(
            keyword,
```

```
            case=False,
            na=False
        )
    ]

    return result[
        [
            "series_id",
            "label",
            "description",
            "link"
        ]
    ]
```

3.1 Commodity Price Indicators

The Bank of Canada publishes several commodity price indexes.

We start with commodity-related indicators because they are important for understanding Canadian economic exposure to natural resources.

[16]:
```
commodity_series = search_series("commodity")

commodity_series.head(20)
```

[16]:
```
              series_id  \
0                 A.AGRI
1                 A.BCNE
2                 A.BCPI
3                 A.ENER
4                 A.FISH
5                 A.FOPR
6                 A.MTLS
4522   FSR_2019_C12_S01
4615    FSR_2020_C11_S1
4725    FSR_2022_C11_S2
4732    FSR_2022_C13_S2
4777    FSR_2022_C3B_S2
5833   INDINF_CPIENER_M
5834     INDINF_CPINE_M
5835    INDINF_CPITOT_M
5846  INDINF_FAM_WIOD_Q
5849     INDINF_GRACE_Q
5971                M.AGRI
5972                M.BCNE
5973                M.BCPI

                                                  label  \
```

```
0                                                  Annual BCPI Agriculture
1                                               Annual BCPI Excluding Energy
2                                                     Annual BCPI Total
3                                                    Annual BCPI Energy
4                                                      Annual BCPI Fish
5                                                  Annual BCPI Forestry
6                                             Annual BCPI Metals and Minerals
4522               Debt at risk, excluding five commodity-related sectors
4615                                           Commodity-related sectors
4725                              Businesses in other commodity industries
4732                              Businesses in other commodity industries
4777   High-yield bonds issued by businesses in other commodity industries
5833                          Bank of Canada commodity price index - Energy
5834                       Bank of Canada commodity price index - Non-energy
5835                            Bank of Canada commodity price index - Total
5846  Foreign demand for Canadian non-commodity exports (FAM-IO) (2007=100)
5849   Foreign demand for Canadian non-commodity exports (GRACE) (2007=100)
5971                                    Monthly BCPI Agriculture - v52673500
5972                                 Monthly BCPI Excluding Energy - v52673497
5973                                       Monthly BCPI Total - v52673496

                                                                 description  \
0                Annual Bank of Canada commodity price index - Agriculture
1             Annual Bank of Canada commodity price index - Excluding Energy
2                      Annual Bank of Canada commodity price index - Total
3                     Annual Bank of Canada commodity price index - Energy
4                       Annual Bank of Canada commodity price index - Fish
5                   Annual Bank of Canada commodity price index - Forestry
6          Annual Bank of Canada commodity price index - Metals and Minerals
4522                                                                     %
4615                                                                   NaN
4725                                                                   NaN
4732                                                                   NaN
4777                                                                   NaN
5833                                       Year-over-year percentage change
5834                                       Year-over-year percentage change
5835                                       Year-over-year percentage change
5846  Foreign demand for Canadian non-commodity exports (FAM-IO) (2007=100)
5849   Foreign demand for Canadian non-commodity exports (GRACE) (2007=100)
5971              Monthly Bank of Canada commodity price index - Agriculture
5972           Monthly Bank of Canada commodity price index - Excluding Energy
5973                    Monthly Bank of Canada commodity price index - Total

                                                    link
0                https://www.bankofcanada.ca/valet/series/A.AGRI
1                https://www.bankofcanada.ca/valet/series/A.BCNE
2                https://www.bankofcanada.ca/valet/series/A.BCPI
```

```
3                https://www.bankofcanada.ca/valet/series/A.ENER
4                https://www.bankofcanada.ca/valet/series/A.FISH
5                https://www.bankofcanada.ca/valet/series/A.FOPR
6                https://www.bankofcanada.ca/valet/series/A.MTLS
4522   https://www.bankofcanada.ca/valet/series/FSR_2019_C12_S01
4615    https://www.bankofcanada.ca/valet/series/FSR_2020_C11_S1
4725    https://www.bankofcanada.ca/valet/series/FSR_2022_C11_S2
4732    https://www.bankofcanada.ca/valet/series/FSR_2022_C13_S2
4777    https://www.bankofcanada.ca/valet/series/FSR_2022_C3B_S2
5833   https://www.bankofcanada.ca/valet/series/INDINF_CPIENER_M
5834     https://www.bankofcanada.ca/valet/series/INDINF_CPINE_M
5835    https://www.bankofcanada.ca/valet/series/INDINF_CPITOT_M
5846  https://www.bankofcanada.ca/valet/series/INDINF_FAM_WIOD_Q
5849     https://www.bankofcanada.ca/valet/series/INDINF_GRACE_Q
5971             https://www.bankofcanada.ca/valet/series/M.AGRI
5972             https://www.bankofcanada.ca/valet/series/M.BCNE
5973             https://www.bankofcanada.ca/valet/series/M.BCPI
```

3.2 Commodity Indicators Observation

The commodity keyword search identifies several groups of indicators.

The most relevant series for the Canadian Financial Lakehouse are the Bank of Canada Commodity Price Index (BCPI) indicators:

- Total commodity index
- Energy index
- Agriculture index
- Metals and minerals index
- Forestry index

The search also returns financial stability indicators related to commodity sectors.

For the first version of the lakehouse, the focus will be on core economic indicators rather than sector-specific financial reports.

4 Financial Domain Exploration

To build a financial data lakehouse, we need to identify the main economic domains available in the Bank of Canada catalog.

We will evaluate several categories:

- Inflation
- Interest rates
- Exchange rates
- Economic growth
- Labour market
- Commodities
- Fixed income markets

[17]:
```
keywords = [
    "inflation",
    "consumer price",
    "interest rate",
    "exchange rate",
    "gross domestic",
    "employment",
    "unemployment",
    "commodity",
    "bond"
]


domain_counts = {}

for keyword in keywords:
    count = search_series(keyword).shape[0]
    domain_counts[keyword] = count


pd.Series(domain_counts).sort_values(
    ascending=False
)
```

[17]:
```
inflation         765
bond              678
consumer price    452
exchange rate     407
gross domestic    292
interest rate     285
employment        258
unemployment      179
commodity          35
dtype: int64
```

4.1 Financial Domain Exploration Results

The keyword-based exploration provides an initial view of the economic domains available in the Bank of Canada catalog.

The largest groups identified are:

- Inflation indicators
- Bond and fixed-income markets
- Consumer price indexes
- Foreign exchange indicators
- Economic activity indicators
- Labour market statistics

These domains represent the main building blocks for a Canadian macro-financial data platform.

The next step is to analyze each domain separately and identify the most relevant time series for the first version of the lakehouse.

5 Inflation Indicators Exploration

Inflation is one of the key economic indicators for financial analysis.

The Bank of Canada provides multiple inflation-related series:

- Consumer Price Index (CPI)
- Core inflation measures
- Inflation expectations
- Year-over-year changes

We explore the available inflation indicators to identify the most relevant candidates.

[18]:
```
inflation_series = search_series("inflation")

inflation_series.head(30)
```

[18]:

	series_id	\
7	ABOVE3	
15	ATOM_V41693242	
278	BELOW1	
958	BOS_BACKGROUNDER_INDICATOR_2026Q2_C3_S3	
970	BOS_BACKGROUNDER_MAGNITUDE_2026Q2_C4_S2	
972	BOS_BACKGROUNDER_MAGNITUDE_2026Q2_C5_S2	
1059	BOS_INFNA_AT	
1060	BOS_INFNA_BC	
1061	BOS_INFNA_CITU	
1062	BOS_INFNA_CPBS	
1063	BOS_INFNA_FIRE	
1064	BOS_INFNA_L	
1065	BOS_INFNA_M	
1066	BOS_INFNA_MANUFACT	
1067	BOS_INFNA_ON	
1068	BOS_INFNA_PR	
1069	BOS_INFNA_PRIMARY	
1070	BOS_INFNA_QC	
1071	BOS_INFNA_S	
1072	BOS_INFNA_TRADE	
1073	BOS_INF_ABOVE_3_AT	
1074	BOS_INF_ABOVE_3_BC	
1075	BOS_INF_ABOVE_3_CITU	
1076	BOS_INF_ABOVE_3_CPBS	
1077	BOS_INF_ABOVE_3_FIRE	
1078	BOS_INF_ABOVE_3_L	

```
1079                       BOS_INF_ABOVE_3_M
1080                  BOS_INF_ABOVE_3_MANUFACT
1081                        BOS_INF_ABOVE_3_ON
1082                        BOS_INF_ABOVE_3_PR

                                                                  label  \
7                                       Above 3 per cent (% of firms)
15                 CPIX, percentage change year-over-year (unadjusted)
278                                     Below 1 per cent (% of firms)
958                                                        CPI inflation
970   IPPI inflation (year-over-year percentage change, right scale)
972    CPI inflation (year-over-year percentage change, right scale)
1059                                                        No response
1060                                                        No response
1061                                                        No response
1062                                                        No response
1063                                                        No response
1064                                                        No response
1065                                                        No response
1066                                                        No response
1067                                                        No response
1068                                                        No response
1069                                                        No response
1070                                                        No response
1071                                                        No response
1072                                                        No response
1073                                                           Above 3%
1074                                                           Above 3%
1075                                                           Above 3%
1076                                                           Above 3%
1077                                                           Above 3%
1078                                                           Above 3%
1079                                                           Above 3%
1080                                                           Above 3%
1081                                                           Above 3%
1082                                                           Above 3%

                                    description  \
7                                                                Inflation
Expectations - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
15    A measure of core inflation that excludes eight of the most volatile
Consumer Price Index (CPI) components - fruit, vegetables, gasoline, fuel oil,
natural gas, mortgage interest, inter-city trans…
278                                                              Inflation
Expectations - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
```

958
CPI inflation
970
IPPI inflation (year-over-year percentage change, right scale)
972
CPI inflation (year-over-year percentage change, right scale)
1059 Inflation expectations of firms in the Atlantic region, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price in…
1060 Inflation expectations of firms in British Columbia, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1061 Inflation expectations of firms in the CITU sector, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1062 Inflation expectations of firms in the CPBS sector, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1063 Inflation expectations of firms in the FIRE sector, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1064 Inflation expectations of large firms, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1065 Inflation expectations of medium-sized firms, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1066 Inflation expectations of firms in the manufacturing sector, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer pri…
1067 Inflation expectations of firms in Ontario, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1068 Inflation expectations of firms in the Prairies region, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price in…
1069 Inflation expectations of firms in the primary sector, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1070 Inflation expectations of firms in Quebec, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1071 Inflation expectations of small firms, four-quarter moving average - Over the next two years, what do you expect the annual rate of inflation to be, based on the consumer price index?
1072 Inflation expectations of firms in the trade sector, four-quarter moving average - Over the next two years, what do you expect the annual rate of

```
inflation to be, based on the consumer price index?
1073  Inflation expectations of firms in the Atlantic region, four-quarter
moving average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price in…
1074    Inflation expectations of firms in British Columbia, four-quarter moving
average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
1075     Inflation expectations of firms in the CITU sector, four-quarter moving
average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
1076     Inflation expectations of firms in the CPBS sector, four-quarter moving
average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
1077     Inflation expectations of firms in the FIRE sector, four-quarter moving
average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
1078                  Inflation expectations of large firms, four-quarter moving
average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
1079           Inflation expectations of medium-sized firms, four-quarter moving
average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
1080  Inflation expectations of firms in the manufacturing sector, four-quarter
moving average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer pri…
1081             Inflation expectations of firms in Ontario, four-quarter moving
average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price index?
1082  Inflation expectations of firms in the Prairies region, four-quarter
moving average - Over the next two years, what do you expect the annual rate of
inflation to be, based on the consumer price in…

  link
7
https://www.bankofcanada.ca/valet/series/ABOVE3
15
https://www.bankofcanada.ca/valet/series/ATOM_V41693242
278
https://www.bankofcanada.ca/valet/series/BELOW1
958
https://www.bankofcanada.ca/valet/series/BOS_BACKGROUNDER_INDICATOR_2026Q2_C3_S3
970
https://www.bankofcanada.ca/valet/series/BOS_BACKGROUNDER_MAGNITUDE_2026Q2_C4_S2
972
https://www.bankofcanada.ca/valet/series/BOS_BACKGROUNDER_MAGNITUDE_2026Q2_C5_S2
1059
https://www.bankofcanada.ca/valet/series/BOS_INFNA_AT
```

1060
https://www.bankofcanada.ca/valet/series/BOS_INFNA_BC
1061
https://www.bankofcanada.ca/valet/series/BOS_INFNA_CITU
1062
https://www.bankofcanada.ca/valet/series/BOS_INFNA_CPBS
1063
https://www.bankofcanada.ca/valet/series/BOS_INFNA_FIRE
1064
https://www.bankofcanada.ca/valet/series/BOS_INFNA_L
1065
https://www.bankofcanada.ca/valet/series/BOS_INFNA_M
1066
https://www.bankofcanada.ca/valet/series/BOS_INFNA_MANUFACT
1067
https://www.bankofcanada.ca/valet/series/BOS_INFNA_ON
1068
https://www.bankofcanada.ca/valet/series/BOS_INFNA_PR
1069
https://www.bankofcanada.ca/valet/series/BOS_INFNA_PRIMARY
1070
https://www.bankofcanada.ca/valet/series/BOS_INFNA_QC
1071
https://www.bankofcanada.ca/valet/series/BOS_INFNA_S
1072
https://www.bankofcanada.ca/valet/series/BOS_INFNA_TRADE
1073
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_AT
1074
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_BC
1075
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_CITU
1076
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_CPBS
1077
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_FIRE
1078
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_L
1079
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_M
1080
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_MANUFACT
1081
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_ON
1082
https://www.bankofcanada.ca/valet/series/BOS_INF_ABOVE_3_PR

5.1 Inflation Indicators Observation

The inflation keyword search reveals multiple groups of indicators:

5.1.1 Core Inflation Measures

Examples:

- CPIX year-over-year percentage change
- Core inflation indicators

These are valuable for macroeconomic analysis.

5.1.2 Inflation Expectations

The catalog contains detailed expectations data:

- By region
- By firm size
- By economic sector

These indicators are useful but may be too granular for the first lakehouse version.

5.1.3 CPI Inflation Indicators

Several series provide headline inflation measures and year-over-year changes.

For the initial implementation, priority will be given to official and widely used inflation indicators with long historical coverage.

5.2 CPI Series Exploration

The Consumer Price Index (CPI) is one of the most important indicators for economic analysis.

Instead of searching all inflation-related concepts, we focus specifically on CPI-related series to identify core candidates.

[19]:
```
cpi_series = df[
    df["search_text"]
    .str.contains(
        "CPI",
        case=False,
        na=False
    )
]


cpi_series[
    [
        "series_id",
        "label",
        "description"
```

```
    ]
].head(30)
```

[19]:

	series_id	label
0	A.AGRI	Annual BCPI Agriculture
1	A.BCNE	Annual BCPI Excluding Energy
2	A.BCPI	Annual BCPI Total
3	A.ENER	Annual BCPI Energy
4	A.FISH	Annual BCPI Fish
5	A.FOPR	Annual BCPI Forestry
6	A.MTLS	Annual BCPI Metals and Minerals
15	ATOM_V41693242	CPIX, percentage change year-over-year (unadjusted)
958	BOS_BACKGROUNDER_INDICATOR_2026Q2_C3_S3	CPI inflation
972	BOS_BACKGROUNDER_MAGNITUDE_2026Q2_C5_S2	CPI inflation (year-over-year percentage change, right scale)
3341	CPIW	CPIW, percentage change year-over-year (unadjusted)
3342	CPI_COMMON	
3343	CPI_MEDIAN	
3344	CPI_TRIM	
5826	INDINF_BOS1TO2_Q	
5827	INDINF_BOS2TO3_Q	
5828	INDINF_BOSBELOW1_Q	
5830	INDINF_BOSOVER3_Q	
5836	INDINF_CPI_COMMON_M	
5837	INDINF_CPI_COMMON_Q	
5838	INDINF_CPI_M	
5839	INDINF_CPI_MEDIAN_M	
5840	INDINF_CPI_MEDIAN_Q	
5841	INDINF_CPI_Q	
5842	INDINF_CPI_TRIM_M	
5843	INDINF_CPI_TRIM_Q	
5844	INDINF_EXPECTSIXTEN_Q	
5845	INDINF_EXPECTTWOTHREE_Q	
5858	INDINF_LOWTARGET	
5874	INDINF_UPPTARGET	

3342	CPI-common
3343	CPI-median
3344	CPI-trim
5826	1 to 2%
5827	2 to 3%
5828	Below 1%
5830	Above 3%
5836	CPI-common
5837	CPI-common
5838	CPI inflation
5839	CPI-median
5840	CPI-median
5841	CPI inflation
5842	CPI-trim
5843	CPI-trim
5844	INDINF_EXPECTSIXTEN_Q
5845	INDINF_EXPECTTWOTHREE_Q
5858	INDINF_LOWTARGET
5874	INDINF_UPPTARGET

	description
0	Annual Bank of Canada commodity price index - Agriculture
1	Annual Bank of Canada commodity price index - Excluding Energy
2	Annual Bank of Canada commodity price index - Total
3	Annual Bank of Canada commodity price index - Energy
4	Annual Bank of Canada commodity price index - Fish
5	Annual Bank of Canada commodity price index - Forestry
6	Annual Bank of Canada commodity price index - Metals and Minerals
15	A measure of core inflation that excludes eight of the most volatile Consumer Price Index (CPI) components - fruit, vegetables, gasoline, fuel oil, natural gas, mortgage interest, inter-city trans…
958	CPI inflation
972	CPI inflation (year-over-year percentage change, right scale)
3341	The year-over-year percentage change in the Consumer Price Index (CPI) with basket weights adjusted inversely to component variability and excluding the effect of changes in indirect taxes.
3342	The measure of common price changes across categories in the Consumer Price

Index (CPI).
3343 The median year-over-year percentage change in the Consumer Price Index (CPI) components, a measure of core inflation published monthly.
3344 A measure of core inflation, published monthly, that excludes components of the Consumer Price Index (CPI) with the most extreme price movements.
5826
Bank of Canada Business Outlook Survey - Percentage of firms expecting CPI inflation over the next two years to be: 1-2%
5827
Bank of Canada Business Outlook Survey - Percentage of firms expecting CPI inflation over the next two years to be: 2-3%
5828
Bank of Canada Business Outlook Survey - Percentage of firms expecting CPI inflation over the next two years to be: Less than 1%
5830
Bank of Canada Business Outlook Survey - Percentage of firms expecting CPI inflation over the next two years to be: More than 3%
5836
The monthly value of the common component of Consumer Price Index (CPI).
5837
The quarterly value of the common component of Consumer Price Index (CPI).
5838
The monthly rate of change in the all-items Consumer Price Index (CPI) for Canada, based on the 2005 basket.
5839 The monthly value of the median Consumer Price Index (CPI). This series measures the central tendency of price changes across CPI components, excluding extreme movements.
5840 The quarterly value of the median Consumer Price Index (CPI). This series measures the central tendency of price changes across CPI components, excluding extreme movements.
5841
The quarterly rate of change in the total Consumer Price Index (CPI).
5842 The monthly value of the trim component of the Consumer Price Index (CPI). This measure excludes CPI components whose rates of change in a given month are located in the tails of the distribution …
5843 The quarterly value of the trim component of the Consumer Price Index (CPI). This measure excludes CPI components whose rates of change in a given month are located in the tails of the distributio…
5844
CPI inflation: Consensus Forecasts - 6-10 years
5845
CPI inflation: Consensus Forecasts - 2-3 years
5858
Consumer Price Index (CPI); Inflation control target, Lower band
5874

```
Consumer Price Index (CPI); Inflation control target, Higher band
```

5.3 CPI Indicators Selection

The CPI exploration identified several important inflation indicators.

The most relevant candidates for the first version of the Canadian Financial Lakehouse are:

5.3.1 Headline Inflation

- `INDINF_CPI_M` - Monthly CPI inflation
- `INDINF_CPI_Q` - Quarterly CPI inflation

5.3.2 Core Inflation Measures

- `CPI_COMMON`
- `CPI_MEDIAN`
- `CPI_TRIM`
- `CPIW`
- `ATOM_V41693242`

These indicators provide complementary views of inflation dynamics:

- Overall price changes
- Core inflation trends
- Measures excluding extreme price movements

The selected indicators will be added to the initial series selection dataset.

6 Create Initial Selected Series Metadata

The objective of this step is to create a curated list of series that will be ingested later.

This selection layer separates:

- Available API catalog
- Business-relevant indicators

The selected series will become the foundation of the Bronze ingestion layer.